

Markov Chains

Definitions, Classification of States, Long-Run Behavior, and Applications

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Motivation

Many real processes evolve in time, and the natural first question is: *how much of the past do we need to remember to predict the future?*

- Full independence (X_{n+1} ignores everything) is usually too strong an assumption.
- Remembering the *entire* history $X_0, \dots, X_n$ is often unnecessary.

Intuition

For many processes (stock prices, weather, a random walk), what happens next depends on *where you are now*, not on the whole path that brought you there. A Markov chain makes this idea precise.

Definition — Markov Chain

Let $\{X_n, n = 0, 1, 2, \dots\}$ be a stochastic process taking values in $\{0, 1, 2, \dots\}$. If $X_n = i$, the process is said to be in **state** i at time n .

The Markov property

$$P\{X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0\} = P_{ij}$$

for all states $i_0, \dots, i_{n-1}, i, j$ and all $n \geq 0$.

In words: the conditional distribution of the **future** given the **past and present** depends *only on the present state* — not on how the process got there. The numbers P_{ij} are the **one-step transition probabilities**.

Basic requirements

$$P_{ij} \geq 0, \quad i, j \geq 0; \quad \sum_{j=0}^{\infty} P_{ij} = 1, \quad i = 0, 1, \dots$$

The Transition Probability Matrix

Collect all the P_{ij} into a matrix $\mathbf{P}$:

$$\mathbf{P} = \begin{pmatrix} P_{00} & P_{01} & P_{02} & \cdots \\ P_{10} & P_{11} & P_{12} & \cdots \\ \vdots & \vdots & \vdots & \\ P_{i0} & P_{i1} & P_{i2} & \cdots \\ \vdots & \vdots & \vdots & \end{pmatrix}$$

- Row i lists the probabilities of jumping *out of* state i .
- Every row sums to 1 — from any state, the chain must go *somewhere* (possibly staying put).
- Once we know the current state and $\mathbf{P}$, the past is irrelevant to predicting the future.

Example — Forecasting the Weather

Suppose the chance of rain tomorrow depends only on whether it is raining **today** — not on any earlier weather. If it rains today, it rains tomorrow with probability α ; if it does not rain today, it rains tomorrow with probability β .

Let state 0 = “rain” and state 1 = “no rain.” This is a two-state Markov chain with

$$\mathbf{P} = \begin{pmatrix} \alpha & 1 - \alpha \\ \beta & 1 - \beta \end{pmatrix}$$

Intuition

Row 0 says: given it rained today, tomorrow it rains again with probability α and clears up with probability $1 - \alpha$. The two entries in each row must add to 1.

Example — A Communications System

Consider a system transmitting digits 0 and 1. Each digit passes through several stages; at each stage there is probability p that the digit is left unchanged.

Letting X_n denote the digit entering stage n , $\{X_n\}$ is a two-state Markov chain with

$$\mathbf{P} = \begin{pmatrix} p & 1-p \\ 1-p & p \end{pmatrix}$$

Intuition

With probability p the bit survives unchanged; with probability $1 - p$ it flips. Only the *current* bit value matters for predicting the next stage.

n -Step Transition Probabilities

We already know the **one-step** probabilities P_{ij} . Define the **n -step transition probability**

$$P_{ij}^n = P\{X_{n+k} = j \mid X_k = i\}, \quad n \geq 0, \quad i, j \geq 0$$

the probability that a process currently in i will be in j after n *additional* transitions. (Of course $P_{ij}^1 = P_{ij}$.)

Note

P_{ij}^n does not depend on the aforementioned variable k due to the the Markov Property

The equations

$$P_{ij}^{n+m} = \sum_{k=0}^{\infty} P_{ik}^n P_{kj}^m \quad \text{for all } n, m \geq 0, \text{ all } i, j$$

Derivation, by conditioning on the state $X_n = k$:

$$\begin{aligned} P_{ij}^{n+m} &= P\{X_{n+m} = j \mid X_0 = i\} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j, X_n = k \mid X_0 = i\} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j \mid X_n = k, X_0 = i\} P\{X_n = k \mid X_0 = i\} \\ &= \sum_{k=0}^{\infty} P_{kj}^m P_{ik}^n \end{aligned}$$

The third line uses the Markov property to drop the conditioning on $X_0 = i$ once we know $X_n = k$.

Chapman–Kolmogorov Equations — Matrix Form

Let $\mathbf{P}^{(n)}$ denote the matrix of n -step probabilities P_{ij}^n . The Chapman–Kolmogorov equations say

$$\mathbf{P}^{(n+m)} = \mathbf{P}^{(n)} \cdot \mathbf{P}^{(m)}$$

where $\cdot$ is ordinary matrix multiplication.

Consequence: powers of $\mathbf{P}$

$$\mathbf{P}^{(2)} = \mathbf{P}^{(1+1)} = \mathbf{P} \cdot \mathbf{P} = \mathbf{P}^2$$

and, by induction,

$$\mathbf{P}^{(n)} = \mathbf{P}^{(n-1+1)} = \mathbf{P}^{n-1} \cdot \mathbf{P} = \mathbf{P}^n$$

So: the n -step transition matrix is just $\mathbf{P}$ multiplied by itself n times.

Example — Four-Day Weather Forecast

Using the weather chain with $\alpha = 0.7$, $\beta = 0.4$:

$$\mathbf{P} = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$$

What is the probability it rains **four days from now**, given it is raining today? We need P_{00}^4 , so we compute $\mathbf{P}^2$, then square it.

$$\mathbf{P}^{(2)} = \mathbf{P}^2 = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix} \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix} = \begin{pmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{pmatrix}$$

$$\mathbf{P}^{(4)} = \left(\mathbf{P}^{(2)}\right)^2 = \begin{pmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{pmatrix} \begin{pmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{pmatrix} = \begin{pmatrix} 0.5749 & 0.4251 \\ 0.5668 & 0.4332 \end{pmatrix}$$

$$P_{00}^4 = 0.5749$$

Definition — Accessible

State j is **accessible** from state i if $P_{ij}^n > 0$ for some $n \geq 0$. Equivalently, starting in i , it is *possible* the process ever enters j .

Definition — Communicate

States i and j that are accessible from each other are said to **communicate**, written $i \leftrightarrow j$.

Intuition

Accessibility is a one-way road: you can get from i to j , but maybe not back. Communication is a two-way road: you can travel between i and j in *both* directions (possibly using different routes and a different number of steps each way).

Communication Is an Equivalence Relation

The relation $\leftrightarrow$ satisfies:

- (i) State i communicates with itself, since $P_{ii}^0 = 1$.
- (ii) If $i \leftrightarrow j$ then $j \leftrightarrow i$ (by symmetry of the definition).
- (iii) If $i \leftrightarrow j$ and $j \leftrightarrow k$ then $i \leftrightarrow k$.

Why (iii) holds

Since $i \leftrightarrow j$ and $j \leftrightarrow k$, there exist n, m with $P_{ij}^n > 0$, $P_{jk}^m > 0$. By Chapman–Kolmogorov,

$$P_{ik}^{n+m} = \sum_{r=0}^{\infty} P_{ir}^n P_{rk}^m \geq P_{ij}^n P_{jk}^m > 0$$

so k is accessible from i ; the reverse direction follows similarly.

Because $\leftrightarrow$ is an equivalence relation, it partitions the state space into disjoint **classes** of mutually communicating states.

Definition

A Markov chain is **irreducible** if there is only *one* class — that is, if *every* pair of states communicates.

Intuition

An irreducible chain is “fully connected” in the communication sense: no matter where you start, you can eventually reach every other state, and from there get back.

Example — Verifying Irreducibility

Consider states 0, 1, 2 with

$$\mathbf{P} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

This chain is irreducible. For instance, to get from state 0 to state 2 (for which $P_{02} = 0$ directly), go

$$0 \rightarrow 1 \rightarrow 2$$

with probability $\frac{1}{2} \cdot \frac{1}{4} > 0$: first from 0 to 1 (probability $\frac{1}{2}$), then from 1 to 2 (probability $\frac{1}{4}$). Checking all other pairs similarly confirms every state communicates with every other.

Example — Classes of a Markov Chain

Consider states 0, 1, 2, 3 with

$$\mathbf{P} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The classes are $\{0, 1\}$, $\{2\}$, and $\{3\}$.

- State 0 (or 1) is accessible from state 2, but *not* vice versa — so $2 \not\leftrightarrow 0$.
- State 3 is **absorbing** ($P_{33} = 1$): once entered, no other state is ever accessible from it.

Example — Rain the Day After Tomorrow: Setup

Recall the four-state rain chain, where whether it rains today depends on the weather over the *last two* days:

- state 0: rained both today and yesterday
- state 1: rained today but not yesterday
- state 2: rained yesterday but not today
- state 3: did not rain either day

$$\mathbf{P} = \begin{pmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{pmatrix}$$

Question: Given that it rained *yesterday and today* (so we are in state 0), what is the probability that it rains the **day after tomorrow**?

Example — Rain the Day After Tomorrow: Solution (1/2)

“Day after tomorrow” is **two** transitions ahead of today, so we need the distribution of X_2 given $X_0 = 0$ — that is, we first need to find the matrix $\mathbf{P}^{(2)} = \mathbf{P} \cdot \mathbf{P}$.

Step 1: multiply $\mathbf{P}$ by itself

$$\begin{pmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{pmatrix} \begin{pmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{pmatrix} = \begin{pmatrix} 0.49 & 0.12 & 0.21 & 0.18 \\ 0.35 & 0.20 & 0.15 & 0.30 \\ 0.20 & 0.12 & 0.20 & 0.48 \\ 0.10 & 0.16 & 0.10 & 0.64 \end{pmatrix}$$

Each entry is a row-times-column dot product. For instance, row 0 against **column 0** and **column 1** of $\mathbf{P}$:

$$P_{00}^{(2)} = 0.7(0.7) + 0(0.5) + 0.3(0) + 0(0) = 0.49$$

$$P_{01}^{(2)} = 0.7(0) + 0(0) + 0.3(0.4) + 0(0.2) = 0.12$$

Example — Rain the Day After Tomorrow: Solution (2/2)

We have now found the matrix

$$\mathbf{P}^{(2)} = \begin{pmatrix} 0.49 & 0.12 & 0.21 & 0.18 \\ 0.35 & 0.20 & 0.15 & 0.30 \\ 0.20 & 0.12 & 0.20 & 0.48 \\ 0.10 & 0.16 & 0.10 & 0.64 \end{pmatrix}$$

Since we start in state 0, we only need **row** 0: (0.49, 0.12, 0.21, 0.18).

Step 2: add $P_{00}^{(2)}$ and $P_{01}^{(2)}$

Rain occurs the day after tomorrow if $X_2 \in \{0, 1\}$ (both states encode “rained that day”), so we add the first two entries of row 0:

$$P_{00}^{(2)} + P_{01}^{(2)} = 0.49 + 0.12 = 0.61$$

Interpretation

Given it rained both yesterday and today, there is a 61% chance it rains again two days from now.

Long-Run Behavior — Limiting Probabilities

As n grows, does P_{ij}^n settle down to a fixed value, no matter where we started (i)? For many chains, yes.

Limiting probabilities

If $\lim_{n \rightarrow \infty} P_{ij}^n$ exists and does **not depend on** i , call this limit $\pi_j = \lim_{n \rightarrow \infty} P_{ij}^n$, $j \geq 0$. Then $\{\pi_j\}$ is the **unique** nonnegative solution of

$$\pi_j = \sum_i \pi_i P_{ij}, \quad j \geq 0$$

$$\sum_j \pi_j = 1$$

Intuition

Once far enough in the future, the chain “forgets” its starting point. The vector $\pi = (\pi_0, \pi_1, \dots)$ is the distribution the chain settles into — and it must be self-consistent: applying one more transition to π should give back π .

Intuition — Balance Equations

The equation $\pi_j = \sum_i \pi_i P_{ij}$ is a **balance equation**:

- π_i = long-run fraction of time spent in state i .
- $\pi_i P_{ij}$ = long-run fraction of transitions that go *from i into j* .
- Summing over all i : the total long-run rate of transitions *into j* .

Equilibrium

In the long run, the rate of transitions **into** state j must equal the fraction of time **spent in j** — otherwise the amount of probability mass sitting in j would be changing over time. This is exactly the “flow in = amount present” condition that pins down a stationary distribution.

Example — Long-Run Weather Proportions: Setup

Return to the weather chain $\mathbf{P} = \begin{pmatrix} \alpha & 1 - \alpha \\ \beta & 1 - \beta \end{pmatrix}$. The balance equations $\pi_j = \sum_i \pi_i P_{ij}$ are just the matrix equation $\pi \mathbf{P} = \pi$, written with π as a row vector:

$$(\pi_0 \quad \pi_1) \begin{pmatrix} \alpha & 1 - \alpha \\ \beta & 1 - \beta \end{pmatrix} = (\pi_0 \quad \pi_1)$$

Expand the left-hand side

$$(\alpha\pi_0 + \beta\pi_1 \quad (1 - \alpha)\pi_0 + (1 - \beta)\pi_1) = (\pi_0 \quad \pi_1)$$

Matching entries on both sides gives the balance equations, together with the normalization condition:

$$\pi_0 = \alpha\pi_0 + \beta\pi_1, \quad \pi_1 = (1 - \alpha)\pi_0 + (1 - \beta)\pi_1, \quad \pi_0 + \pi_1 = 1$$

We only need two of these three equations (one balance equation is always redundant once we include $\sum_j \pi_j = 1$).

Example — Long-Run Weather Proportions: Solving

From $\pi_0 = \alpha\pi_0 + \beta\pi_1$:

$$\pi_0 - \alpha\pi_0 = \beta\pi_1 \implies \pi_0(1 - \alpha) = \beta\pi_1 \implies \pi_1 = \frac{(1 - \alpha)}{\beta}\pi_0$$

Substitute into $\pi_0 + \pi_1 = 1$:

$$\pi_0 + \frac{1 - \alpha}{\beta}\pi_0 = 1 \implies \pi_0 \left(\frac{\beta + 1 - \alpha}{\beta} \right) = 1$$

$$\boxed{\pi_0 = \frac{\beta}{1 + \beta - \alpha}},$$

$$\boxed{\pi_1 = 1 - \pi_0 = \frac{1 - \alpha}{1 + \beta - \alpha}}$$

Plugging in numbers

With $\alpha = 0.7$, $\beta = 0.4$:

$$\pi_0 = \frac{0.4}{1 + 0.4 - 0.7} = \frac{0.4}{0.7} = \frac{4}{7} \approx 0.571$$

So in the long run it rains about 57.1% of days.

Two Properties of Markov Chains: Periodicity

Property 1 — Period

State i has **period** d if $P_{ii}^n = 0$ whenever n is **not** divisible by d , and d is the *largest* integer with this property.

Property 2 — Aperiodic

A state with period 1 is said to be **aperiodic**.

Intuition

The period is the “rhythm” of return visits: if $d = 2$, the chain can only ever get back to i after an even number of steps — it is forced to alternate between two “sides” before it can come home.

Periodic Chains Have No Limiting Probabilities

Consider the two-state chain with $P_{01} = P_{10} = 1$ (it alternates states every step). Here $d = 2$: the chain can only return to state 0 after an even number of steps.

Long-run proportions exist...

Since the chain spends exactly half its time in each state, $\pi_0 = \pi_1 = \frac{1}{2}$.

...but limiting probabilities do not

$$P_{00}^n = \begin{cases} 1 & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

which never settles down as $n \rightarrow \infty$.

Takeaway

A periodic chain does not have a limiting probability — even though the long-run *proportions* of time in each state are still well defined. Limiting probabilities require the chain to be irreducible *and aperiodic*.

Example — The Hardy–Weinberg Law: Setup

In a large population, each individual has a gene pair of type AA , aa , or Aa , with respective proportions p_0, q_0, r_0 (where $p_0 + q_0 + r_0 = 1$). Mating is random: each parent contributes one randomly chosen gene.

Intuition

Choosing a random gene from a random parent is the same as choosing a random gene from the *entire* gene pool. So we first find the probability a randomly drawn gene is type A or type a .

Conditioning on the parent's gene pair:

$$P\{A\} = P\{A \mid AA\}p_0 + P\{A \mid aa\}q_0 + P\{A \mid Aa\}r_0 = p_0 + \frac{r_0}{2}$$

Similarly, $P\{a\} = q_0 + \frac{r_0}{2}$.

Example — Hardy–Weinberg: Next-Generation Proportions

A child's two genes are drawn independently from the gene pool, so the child's type is determined by two independent draws with probabilities $P\{A\}$ and $P\{a\}$ above. Writing p, q, r for the next generation's proportions of AA, aa, Aa :

$$p = P\{A\}P\{A\} = \left(p_0 + \frac{r_0}{2}\right)^2$$

$$q = P\{a\}P\{a\} = \left(q_0 + \frac{r_0}{2}\right)^2$$

$$r = 2 P\{A\}P\{a\} = 2 \left(p_0 + \frac{r_0}{2}\right) \left(q_0 + \frac{r_0}{2}\right)$$

(the factor 2 for r because Aa can arise as “ A from mother, a from father” or “ a from mother, A from father”).

Example — Hardy–Weinberg: The Gene Pool Is Conserved

Does the fraction of type- A genes change from generation to generation? By the same reasoning as before, a randomly chosen gene from the *new* generation is type A if the individual is AA , or is Aa and happens to pass on its A gene (half the time). So

$$P\{A\} = p + \frac{r}{2}$$

Substituting the formulas for p and r found earlier:

$$\begin{aligned} P\{A\} &= p + \frac{r}{2} = \left(p_0 + \frac{r_0}{2}\right)^2 + \left(p_0 + \frac{r_0}{2}\right) \left(q_0 + \frac{r_0}{2}\right) \\ &= \left(p_0 + \frac{r_0}{2}\right) \left[p_0 + q_0 + r_0\right] && \text{(factor; } p_0 + q_0 + r_0 = 1) \\ &= p_0 + \frac{r_0}{2} \end{aligned}$$

Hardy–Weinberg Law

So $P\{A\} = p + r/2$ is unchanged from the previous generation! By the exact same argument, $P\{a\} = q + r/2$ is unchanged as well. The proportions p, q, r therefore stay **fixed** in every generation after the first.

Example — Hardy–Weinberg: A Markov Chain of Descendants

Now follow a *single* individual's lineage (one offspring per generation), once proportions have stabilized at p, q, r . The genetic type of generation n , X_n , is a Markov chain on $\{AA, aa, Aa\}$, with rows and columns ordered AA, aa, Aa :

$$\mathbf{P} = \begin{pmatrix} p + \frac{r}{2} & 0 & q + \frac{r}{2} \\ 0 & q + \frac{r}{2} & p + \frac{r}{2} \\ \frac{p}{2} + \frac{r}{4} & \frac{q}{2} + \frac{r}{4} & \frac{p}{2} + \frac{q}{2} + \frac{r}{2} \end{pmatrix}$$

Intuition

Each row is obtained by conditioning on the state of the randomly chosen mate — e.g., an AA parent contributes an A gene for sure, so the child is AA with probability

$$P\{A \text{ from mate}\} = p + r/2.$$

Example — Hardy–Weinberg: Stationary Distribution

It is intuitive that the limiting (stationary) probabilities for this chain are exactly p, q, r — the same fractions found across the whole population. To verify, we check $\pi_{AA} = p$ satisfies the balance equation (one equation is redundant, so two checks suffice):

$$\begin{aligned}p &\stackrel{?}{=} p \left(p + \frac{r}{2} \right) + r \left(\frac{p}{2} + \frac{r}{4} \right) = \left(p + \frac{r}{2} \right)^2 \\q &\stackrel{?}{=} q \left(q + \frac{r}{2} \right) + r \left(\frac{q}{2} + \frac{r}{4} \right) = \left(q + \frac{r}{2} \right)^2\end{aligned}$$

But we already showed $p + r/2 = P\{A\}$, and p is *defined* as $P\{A\}^2$ — so $p = (p + r/2)^2$ holds automatically! The same argument gives the q equation. Hence $\pi_{AA} = p$, $\pi_{aa} = q$, $\pi_{Aa} = r$ is indeed the stationary distribution.

The Gambler's Ruin Problem — Setup

A gambler wins one unit with probability p and loses one unit with probability $q = 1 - p$ on each play, independently. The gambler stops when reaching a fortune of N (success) or 0 (ruin).

Let X_n be the fortune at time n : a Markov chain with

$$P_{00} = P_{NN} = 1, \quad P_{i,i+1} = p = 1 - P_{i,i-1}, \quad i = 1, \dots, N-1$$

Intuition

States 0 and N are **absorbing** walls; every state in between is transient (visited only finitely often). So with probability 1 the gambler eventually hits 0 or N — the question is: *which one, and with what probability?*

Let $P_i = P\{\text{fortune eventually reaches } N \mid \text{start at } i\}$.

Gambler's Ruin — Setting Up the Recursion

Condition on the outcome of the first play:

$$P_i = pP_{i+1} + qP_{i-1}, \quad i = 1, \dots, N-1$$

Since $p + q = 1$, write $pP_i + qP_i = pP_{i+1} + qP_{i-1}$, which rearranges to

$$p(P_{i+1} - P_i) = q(P_i - P_{i-1})$$

$$\boxed{P_{i+1} - P_i = \frac{q}{p} (P_i - P_{i-1})}, \quad i = 1, \dots, N-1$$

Intuition

This says: the “jump” in win-probability from i to $i+1$ is a fixed multiple q/p of the previous jump. That is exactly the signature of a **geometric** sequence of differences.

Gambler's Ruin — Telescoping the Differences

Since $P_0 = 0$, apply the boxed recursion repeatedly, starting from $P_1 - P_0 = P_1$:

$$P_2 - P_1 = \frac{q}{p}(P_1 - P_0) = \frac{q}{p}P_1$$

$$P_3 - P_2 = \frac{q}{p}(P_2 - P_1) = \left(\frac{q}{p}\right)^2 P_1$$

$$\vdots$$

$$P_i - P_{i-1} = \frac{q}{p}(P_{i-1} - P_{i-2}) = \left(\frac{q}{p}\right)^{i-1} P_1$$

$$\vdots$$

$$P_N - P_{N-1} = \left(\frac{q}{p}\right)^{N-1} P_1$$

Each difference is the *previous* difference scaled by q/p — a geometric progression with first term P_1 and ratio q/p .

Gambler's Ruin — Summing to Solve for P_i

Add the first $i - 1$ of the telescoping equations; the left side collapses to $P_i - P_1$:

$$P_i - P_1 = P_1 \left[\frac{q}{p} + \left(\frac{q}{p} \right)^2 + \cdots + \left(\frac{q}{p} \right)^{i-1} \right]$$

Summing the finite geometric series (ratio q/p) gives

$$P_i = \begin{cases} \frac{1 - (q/p)^i}{1 - (q/p)} P_1, & \frac{q}{p} \neq 1 \\ i P_1, & \frac{q}{p} = 1 \end{cases}$$

Intuition

P_i is built entirely from P_1 : once we know the probability of winning starting one unit from ruin, every other P_i follows from the fixed geometric ratio q/p .

Gambler's Ruin — Solving for P_1

We still need P_1 . Use the boundary condition $P_N = 1$ (reaching N from N is certain):

$$1 = P_N = \begin{cases} \frac{1 - (q/p)^N}{1 - (q/p)} P_1, & p \neq \frac{1}{2} \\ N P_1, & p = \frac{1}{2} \end{cases} \implies P_1 = \begin{cases} \frac{1 - (q/p)}{1 - (q/p)^N}, & p \neq \frac{1}{2} \\ \frac{1}{N}, & p = \frac{1}{2} \end{cases}$$

Substituting back into the formula for P_i :

Gambler's Ruin Formula

$$P_i = \begin{cases} \frac{1 - (q/p)^i}{1 - (q/p)^N}, & p \neq \frac{1}{2} \\ \frac{i}{N}, & p = \frac{1}{2} \end{cases}$$

Gambler's Ruin — What Happens as $N \rightarrow \infty$?

Imagine playing against an infinitely rich adversary ($N \rightarrow \infty$):

$$P_i \longrightarrow \begin{cases} 1 - \left(\frac{q}{p}\right)^i, & p > \frac{1}{2} \\ 0, & p \leq \frac{1}{2} \end{cases}$$

Intuition

- If the game favors you ($p > \frac{1}{2}$), there is a genuine positive chance your fortune grows without bound forever.
- If the game is fair or unfavorable ($p \leq \frac{1}{2}$), you are **certain** to go broke eventually against an opponent with unlimited capital — even a fair game is not safe in the long run.

Numerical Illustration

Patty and Max flip pennies; Patty wins each flip with probability $p = 0.6$. Patty starts with 5 pennies, Max with 10 (so $N = 15$). What is the probability Patty wipes Max out?

Here $q/p = 0.4/0.6 = 2/3$, $i = 5$, $N = 15$:

$$P_5 = \frac{1 - (2/3)^5}{1 - (2/3)^{15}} = \frac{1 - 0.1317}{1 - 0.0023} = \frac{0.8683}{0.9977} \approx 0.87$$

Intuition

Even with a modest edge (60% vs. 40% per flip), Patty's win probability of 87% is far higher than her share of the total pennies ($5/15 \approx 33\%$) — small per-play edges compound strongly over a long sequence of plays.